Alexander Lu

+1(858)-688-4567 | alexander.lu@berkeley.edu | linkedin.com/in/alexanderlu1258 | github.com/YLu-1258 | www.alexlu.us

Education

University of California, Berkeley Aug. 2024 – May 2028
B.S. Electrical Engineering and Computer Science, Mathematics Minor *GPA: 4.00/4.00*

- Courses: Data Structures, Algorithms, Computer Architecture, Linear Algebra, Discrete Math, Stochastic Processes, Signal Processing, Real Analysis, Abstract Algebra, Circuit Analysis, Deep Neural Networks, Compilers

Publications & Research

Jan. 2026 – **Evolutionary Strategies Lead to Catastrophic Forgetting in LLMs** - ACL 2026 Main Conference
Jun. 2026 – **Automated Discovery Has No Universally Superior Harness** - COLM 2026 LM4Sci Workshop
Aug. 2026 – **Inference-Time Sampling in Language Models** - Manuscript in preparation
Aug. 2026 – **Continual Adaptation of Language Models at Deployment** - Ongoing research

Experience

Berkeley AI Research – Berkeley Speech Group May 2025 – Present
Machine Learning Researcher

- Post-trained Qwen3 and Llama reasoning models using SFT, DPO, GRPO, and Evolutionary Strategies to improve chain-of-thought faithfulness and exploitative poker performance.
- Investigated catastrophic forgetting in LLM post-training, finding Evolutionary Strategies induce denser weight updates and higher ℓ_2 norms than GRPO; work accepted to **ACL 2026**.
- Evaluated **30** LLM discovery harnesses across **12** model-problem pairs and **3.1M+** LLM rollouts, showing no fixed harness generalizes universally and developing adaptive compute allocation; accepted to **COLM 2026 LM4Sci** Workshop.

Apple – Robotics and Applied Intelligence Apr. 2026 – Aug. 2026
Applied AI SWE Intern

- Developing a statistical heatmap system to identify regressions and improvements across **1000+** hardware and software builds, enabling faster comparison of multi-product test results.
- Building an unsupervised anomaly-detection and root-cause analysis pipeline for time-series data collected from **500+** devices.
- Deployed an RL policy from NVIDIA Isaac Sim to a Franka manipulator and diagnosed unstable Sim2Real behavior caused by mismatched controller frequencies using custom policy-output visualizations.

Amazon – Traffic Engineering Jan. 2026 – Apr. 2026
Software Development Engineering Intern

- Built an agentic AI orchestration on AWS to automate CDN deployments across **20+** global regions.
- Enabled safe, human-in-the-loop traffic shifts at **300K+ TPM** in under **10 minutes** with GenAI monitoring and validation.
- Reduced deployment time by **59%** and cut potential human error points by **82%**, improving production reliability.

UCSF Weill Institute for Neurosciences – Chen Lab Jan. 2025 – Dec. 2025
Machine Learning Intern

- Employed signal processing methods (Wavelet Transform, Fourier Analysis) with SciPy for EEG feature extraction.
- Engineered seizure detection pipeline with EEG time series using SVM and GMM models; achieved **95%** accuracy.
- Trained a Keypoint MoSeq pipeline to perform unsupervised learning on detecting rodent movements.
- Optimized memory usage in a seizure-detection Python library from **200GB+** down to **4GB** via chunking.

Projects

EigenAgent | github.com/YLu-1258/eigenAgent

- Built a fully local LLM desktop assistant using Rust and llama.cpp for offline, on-device inference with GGUF-quantized models.
- Implemented token-level streaming via an event-driven pipeline, improving responsiveness and reducing latency.
- Integrated MCP-based tool invocation to support agentic workflows and extensible custom tool servers.

Semantic Backlinks Generator | GitHub

- Python backlink generator for Obsidian-style blogs using a sentence transformer for semantic encoding.
- Employed FAISS for efficient similarity search to surface semantically related posts and reduce manual linking effort.
- Built a data-visualization pipeline for customizing density, confidence level, and semantic weights on a graph.

Mechanistic Interpretability Toolkit | GitHub

- Engineered a mechanistic interpretability toolkit for GPT-2 and Qwen3 supporting token attribution, attention head analysis, and activation patching via TransformerLens, FastAPI, and React.js.
- Implemented activation patching across 28 layers and 24 heads to identify causal circuits with real-time ΔLogit heatmap visualization.
- Streamed layer-wise logit lens predictions via WebSocket, visualizing the transformer layers where next-token predictions stabilize.

Skills

Languages: C/C++, Java, Python, Rust, TypeScript/HTML/CSS, SQL, Swift
Frameworks & Tools: NumPy, Pandas, PyTorch, TensorFlow, Scikit-Learn, VERL, vLLM, SciPy, FAISS, React.js, Flask, Spring Boot, Node.js, Docker, AWS, Git, Linux
Concepts: Reinforcement Learning, LLM Training, Distributed Systems, Time-Series Analysis, Signal Processing